SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DIAGNOSIS AND PHENOTYPE CLASSIFICATION;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DIAGNOSIS AND PHENOTYPE CLASSIFICATION;BEDSIDE DIAGNOSTIC CRITERIA AND ANTHROPOMETRIC MEASUREMENT;
For children 6–59 months, diagnose ##SAM## at bedside using any one of three entry criteria: ##MUAC## <115 mm, severe wasting by ##WHZ##/##WFH## (or ##WLZ##) <−3, or bilateral pitting nutritional oedema. Indian guidance from ##MoHFW## and ##IAP## uses these same markers and treats them as pathway-defining rather than merely descriptive, because phenotype and complications determine disposition, fluid strategy, and feeding volumes.

For program entry and clinical safety, use whichever criterion is positive and do not exclude a child because only one indicator meets threshold. ##MUAC## and ##WHZ## identify overlapping but non-identical high-risk groups; therefore, document both at baseline whenever feasible and avoid discharge decisions based on a single normalized marker if another remains severely abnormal.

Separate “admission/eligibility” from “prescription and monitoring.” Weight is required to prescribe therapeutic food quantities and inpatient feed volumes, but oedema can mask wasting and distort early weight change. During stabilization, track daily clinical status (appetite, temperature, mental state), oedema grade, and intake volumes; use serial weight gain only after oedema is clearly reducing and the child is on catch-up feeding. Infants under 6 months require a feeding-effectiveness pathway rather than these same thresholds and are addressed separately.

When anthropometry is used, minimize measurement error: use a calibrated scale, measure length/height with a board, and take ##MUAC## at the mid-point between acromion and olecranon on the left arm with the tape snug but not tight. Repeat and record the most plausible value, because small errors near cut-offs can change triage decisions.
References:
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DIAGNOSIS AND PHENOTYPE CLASSIFICATION;RECOGNITION AND DIFFERENTIAL OF BILATERAL PITTING OEDEMA;

Classify ##SAM## as non-oedematous (severe wasting) versus oedematous (nutritional oedema, classically termed kwashiorkor) at first contact, because oedema changes fluid targets, feeding volumes, monitoring priorities, and disposition risk. Confirm oedema clinically: press a thumb gently for about 10 seconds on the dorsum of each foot; a persistent pit that is bilateral indicates nutritional oedema. Grade severity as + (feet only), ++ (feet plus lower legs and/or hands/forearms), and +++ (generalized, including face). Oedema can make a child appear “not too thin” and can falsely elevate weight, so early weight loss during treatment may represent clinical improvement from oedema resolution rather than deterioration.

Do not treat nutritional oedema with diuretics. Oedema in ##SAM## reflects systemic metabolic and electrolyte derangements (not isolated fluid excess), and sodium loads or diuresis can precipitate shock or cardiac failure. Use oedema presence to raise infection vigilance, since oedematous phenotypes in Indian facility data have higher inpatient mortality than nonoedematous cases.

Actively exclude alternative causes when the pattern is atypical: unilateral oedema, non-pitting oedema, hypertension, gross ascites out of proportion to pedal oedema, or persistent oedema despite correct ##SAM## care should prompt evaluation for nephrotic syndrome, chronic liver disease, or heart failure. A focused bedside screen is urinalysis for protein, blood pressure, cardiac examination, and review of sodium exposures (salty foods, high-sodium ##IV## fluids, incorrect feed recipes).

References:

- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_ FBCSA-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;PHYSIOLOGIC ADAPTATIONS THAT ALTER FLUID AND FEEDING DECISIONS;

Severe malnutrition produces a state of physiologic downregulation (“reductive adaptation”) with reduced cardiac, renal, hepatic, and immune reserve. Clinically, children may not mount fever or tachycardia despite invasive infection, and they tolerate salt and water loads poorly. This is why complicated ##SAM## should be treated as high risk even when signs appear “mild,” and why standard pediatric bolus fluids, high-sodium feeds, and early catch-up feeding can precipitate heart failure.

Infection can look “quiet” in ##SAM##. Hypothermia or recurrent hypoglycemia should be managed as sepsis-equivalent warning signs until proven otherwise, with empiric antibiotics and close monitoring. Laboratory values may also be deceptively reassuring; interpretation should be driven by clinical syndrome and response, not by “normal” inflammatory markers.

Nutritional edema is a metabolic state combining potassium depletion and sodium retention; total body sodium may be high even when plasma sodium is low. Therefore, do not use

diuretics for edema and hard-code sodium restriction in feeds and fluids. Intravascular depletion can coexist with edema, so edema must not be used to rule out dehydration.

Total-body potassium and magnesium deficits are universal and take weeks to correct. Early management must include routine potassium and magnesium repletion (rather than waiting for lab confirmation), slow rehydration with frequent reassessment, and staged feeding (##F-75## stabilization before rehabilitation). These physiologic principles explain the core safety rules of inpatient and outpatient ##SAM## protocols.

References:

- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;TRIAGE AND DISPOSITION BETWEEN INPATIENT AND CMAM PATHWAYS;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;TRIAGE AND DISPOSITION BETWEEN INPATIENT AND CMAM PATHWAYS;COMPLICATED VERSUS UNCOMPLICATED SAM DISPOSITION CRITERIA;

The key decision in ##SAM## management is whether the child is “complicated” and needs inpatient stabilization or “uncomplicated” and can be treated in an outpatient ##CMAM## pathway. Make this decision immediately after a rapid danger-sign screen, and treat emergency signs before completing full anthropometry.

Define complicated ##SAM## as meeting a ##SAM## diagnostic criterion plus any medical complication or a failed appetite assessment. Complications that should trigger admission include altered mental state or convulsions, hypoglycemia, hypothermia, fever ≥39°C, persistent vomiting, respiratory distress, suspected severe dehydration or shock, severe palmar pallor or suspected severe anemia, extensive skin infection needing parenteral therapy, or any clinician concern that home care is not feasible.

Oedema increases risk and changes placement thresholds. Severe oedema (+++) is an automatic inpatient indication in Indian protocols. The 2023 ##WHO## guideline allows trained community teams to treat selected children with nutritional oedema outside hospital, but outpatient system reliability varies across India. A safe default for trainees is to admit any oedematous ##SAM## unless a well-supervised outpatient program explicitly includes oedema monitoring and rapid referral.

Define uncomplicated ##SAM## as a clinically stable child with no danger signs and a passed appetite test, with a caregiver able to return reliably and deliver frequent feeds safely. Treat

caregiver barriers (distance, inability to administer feeds, unsafe hygiene) as practical complications that justify inpatient initiation.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_f b mc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition

— https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;TRIAGE AND DISPOSITION BETWEEN INPATIENT AND CMAM PATHWAYS;APPETITE TEST PROTOCOL FOR OUTPATIENT ELIGIBILITY;

Use the appetite test as a triage gate for outpatient eligibility and as an early marker of metabolic stability in ##SAM##. A “pass” means the child is clinically stable enough to accept therapeutic feeding at home with reliable follow-up; a “fail” should be treated as a medical complication requiring inpatient stabilization, not as poor cooperation.

- Practical execution: ensure the child has not had a substantial feed for about 2 hours; conduct the test in a calm area with the caregiver present; offer small, frequent tastes without force; and observe swallowing, endurance, and willingness. Use the same therapeutic product that will be prescribed (standard ##RUTF## where available, or the locally prepared therapeutic feed used by the program) and document the amount consumed.
- Interpretation should be objective. Failure includes inability to consume the minimum program amount for weight band, repeated vomiting during the test, lethargy or worsening respiratory distress while feeding, or caregiver inability to safely administer and store the product. Treat failure as a trigger to start the inpatient “first hour” actions (warmth, glucose prevention, empiric antibiotics if complicated) and initiate supervised stabilization feeding. - Re-test only after stabilization. If hypoglycemia, hypothermia, or sepsis is suspected, the priority is treatment and frequent small feeds; repeating the appetite test before the child is warm, alert, and clinically improving can misclassify severity and delay admission.

References:

- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;TRIAGE AND DISPOSITION BETWEEN INPATIENT AND CMAM PATHWAYS;ADMISSION CRITERIA FOR INFANTS UNDER 6 MONTHS WITH ACUTE MALNUTRITION;

Do not apply the standard ##MUAC## thresholds or ##RUTF## appetite-test logic to infants under 6 months. Disposition is driven by feeding effectiveness and physiologic stability in the mother–infant dyad, because small absolute intake deficits can cause rapid decompensation. - Admit an infant <6 months with acute malnutrition if any danger sign is present (lethargy, convulsions, hypothermia, hypoglycemia, respiratory distress, persistent vomiting), if there is any bilateral pitting edema, or if the infant is too weak to breastfeed or suck effectively. Consider admission when there is recent weight loss or failure to gain, ineffective attachment, or clear caregiver inability to provide safe, frequent feeding at home.

- For anthropometry, use weight-for-length <−3 ##SD## when length is reliably measured (commonly when length is >45 cm). When length measurement is unreliable in very short or sick infants, visible severe wasting, failure to gain, and feeding failure are safer clinical triggers than a single calculated z score.
- Outpatient care is appropriate only when feeding is effective on direct observation: the infant attaches and sucks well, the caregiver can demonstrate frequent feeding (including expressed breast milk if needed), there is no edema, and follow-up can occur at least weekly with rapid referral capability.
- Initial inpatient priorities remain the same as older children: prevent fasting, maintain warmth, correct hypoglycemia, and treat suspected infection early. Transition rapidly to lactation support, observed breastfeeding/expressed breast milk plans, and structured reassessment of feeding effectiveness before considering discharge. References:
- World Health Organization, Severe acute malnutrition in infants (ELENA guidance) — https://www.who.int/tools/elena/interventions/sam-infants
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;STABILIZATION AND REHABILITATION TEN-STEP CARE MODEL;

Use the classic ten-step ##WHO## inpatient care model as a single mental checklist, because early deaths in ##SAM## are often iatrogenic (fluid and feeding errors) and danger signs may be muted. The same steps also structure outpatient ##CMAM## care, but “where” they occur

depends on whether disease is complicated (inpatient stabilization) or uncomplicated (outpatient therapeutic feeding with rapid referral capability). The ten steps are:

1. Treat or prevent hypoglycemia.
2. Treat or prevent hypothermia.
3. Treat or prevent dehydration (no ##IV## unless shock).
4. Correct electrolyte imbalance (potassium and magnesium; restrict sodium).
5. Treat infection empirically (signs may be absent).
6. Correct micronutrient deficiencies (delay iron until rehabilitation).
7. Start cautious feeding with ##F-75## (small, frequent feeds; avoid rapid escalation).
8. Achieve catch-up growth with ##F-100## and/or ##RUTF## as tolerated.
9. Provide sensory stimulation and emotional support to the child and caregiver.
10. Prepare for follow-up after recovery (link to community services and relapse prevention). Anchor these steps to ward time windows to support decision-making: day 0–2 focuses on steps 1–7 (stabilization and prevention of early deaths); day 3–7 focuses on safe transition into step 8 while continuing infection and electrolyte care; week 2 onward focuses on sustained catch-up growth, developmental support, and a structured handover to outpatient follow-up with explicit non-response triggers. References:

- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;CLINICALLY EXECUTABLE ARRIVAL ASSESSMENT AND TRIAGE ORDER SET;

The first 24–72 hours are the highest-risk period in inpatient ##SAM## care. Avoidable deaths commonly follow missed danger signs, delayed glucose/warmth/feeding actions, or iatrogenic “standard pediatrics” fluids and feeds that do not respect the altered physiology of severe malnutrition.

Immediate sequence on arrival: perform an ##ABCD## emergency screen while simultaneously checking bedside glucose and temperature. Treat hypoglycemia immediately when suspected or confirmed, start active warming for hypothermia, and give oxygen for hypoxia or severe respiratory distress. If the child meets criteria for complicated ##SAM##, start empiric antibiotics early because fever may be absent despite bacteremia.

Focused history that changes decisions should be obtained in parallel with stabilization: last feed and current intake; breastfeeding status; vomiting; watery versus bloody diarrhea; urine output; cough or difficult breathing; recent measles exposure; contact with ##TB##; and prior antibiotics. These map directly to syndrome pathways (pneumonia, dysentery, ##UTI##, malaria, ##TB##) and to escalation planning.

Targeted examination: document anthropometry (weight, length/height, ##MUAC##, ##WHZ##/##WLZ##) and edema grade; check for pallor and heart failure signs; look actively for localizing infection (lungs, ears, skin, mouth ulcers, kwashiorkor dermatosis); and reassess dehydration cautiously, recognizing that edema does not exclude intravascular depletion.

First-day monitoring should be predefined: track temperature, respiratory rate, pulse, mental state, intake offered/taken, vomits/stools, and urine frequency. During any rehydration, reassess every 30 minutes for the first 2 hours then hourly, and stop fluids if respiratory rate or pulse rise from baseline or edema worsens. Start ##F-75## as soon as feasible and avoid prolonged fasting.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;FIRST HOUR BUNDLE AND STABILIZATION FEEDING ORDERS;

In inpatient care, execute a “first hour” bundle before detailed workup. Triage immediately for any child suspected to have ##SAM## and do not wait for complete measurements to start lifesaving actions. Prevent and treat hypoglycemia at arrival: check glucose, but treat immediately if lethargic, unconscious, or convulsing. A pragmatic threshold used in Indian protocols is <54 mg/dL: give 10% dextrose 5 mL/kg by ##IV## bolus for severe symptoms, then give oral/##NG## glucose or sucrose and recheck in 30 minutes. Start feeds early and maintain 2-hourly feeding day and night to avoid fasting.

Prevent and treat hypothermia concurrently. If axillary temperature is <35°C, initiate active warming (kangaroo mother care, caps and socks, blankets, warm room) and feed frequently. Assume infection and start empiric antibiotics promptly; fever may be absent despite serious bacterial infection in ##SAM##.
Write stabilization “do-not-do” rules into admission orders because many early deaths are iatrogenic. Do not give large-volume ##IV## fluids unless shock criteria are met. Do not give diuretics to reduce nutritional oedema. Do not start iron in the stabilization phase. Avoid highsodium feeds and salty family foods.
Start feeding immediately with ##F-75## using small, frequent feeds every 2–3 hours. A practical total fluid target is about 130 mL/kg/day, reduced to about 100 mL/kg/day if severe oedema is present. If the child takes ≤80% of the prescribed volume for two consecutive feeds, place a ##NG## tube and give the remaining volume rather than skipping feeds. Do not administer ##RUTF## via ##NG## tube.
References:
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;STANDARDIZED ADMISSION ORDERS AND MONITORING CHART FIELDS;
Standardized ward artifacts reduce omissions and prevent iatrogenic harm in the first 24–72 hours of inpatient ##SAM## care. Use a pre-printed or electronic admission order set and a minimum monitoring chart so that feeding, warming, and antibiotic steps occur without delay.

Admission order set core elements should be written as actionable orders: admit to warm area; check bedside glucose and treat if <54 mg/dL or symptomatic; start active warming if hypothermic; start ##F-75## immediately as per weight-based chart (2-hourly day and night); start empiric antibiotics if complicated; prescribe routine potassium and magnesium with sodium restriction; order minimum labs (glucose, hemoglobin if pallor, electrolytes including phosphate when available, malaria test by geography, and cultures where feasible without delaying antibiotics); document anthropometry (weight, length/height, ##MUAC##, ##WHZ##/##WLZ##) and edema grade.

The monitoring chart should capture the variables that signal silent deterioration: time-stamped temperature, pulse, respiratory rate, mental status, oxygen saturation when available, feed offered and taken at each feed, vomits and stools, urine output proxy, edema grade, and explicit overhydration/heart failure signals (rising respiratory rate and pulse trends, increasing liver size, new periorbital puffiness).

Escalation rules must be printed on the chart: intake <80% of prescribed volume, recurrent hypoglycemia or hypothermia, rising respiratory rate or pulse during rehydration or feed advancement, and failure-to-respond timepoints (for example, appetite not returning by day 4) should trigger senior review and diagnostic reassessment rather than continuation of the same plan.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
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SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;MINIMUM INVESTIGATION SET AND HIDDEN INFECTION REASSESSMENT TRIGGERS;

Minimum investigations in ##SAM## should be taught as “always if available” versus “conditional if non-response,” because invasive infection and metabolic derangement can present without fever or strong localizing signs. Order tests that change immediate bedside actions first, then add targeted tests when recovery stalls.

On arrival or day 0 (where feasible), prioritize point-of-care glucose and a hemoglobin (or ##PCV##), because hypoglycemia and severe anemia drive early mortality and alter fluid and transfusion decisions. Add malaria testing in endemic geographies or when fever is present, and obtain a blood culture before antibiotics only if it will not delay empiric therapy. If resources allow, include serum electrolytes with phosphate during transition feeding, since hypophosphatemia can signal refeeding risk.

During non-response or relapse of instability, actively search for “hidden infection” and comorbidity. A practical escalation bundle is urine microscopy (and culture where feasible) for overlooked urinary infection, stool examination for persistent diarrhea or giardiasis, and chest radiography when chronic pneumonia or ##TB## is suspected. Consider targeted screening for ##HIV## and ##TB## per national algorithms when weight gain is poor despite verified therapeutic feeding.

Resource-adapted rule: if labs are unavailable, treat patterns such as recurrent hypoglycemia or hypothermia, persistent anorexia beyond expected timepoints, oedema not reducing, or weight gain <5 g/kg/day after catch-up initiation as mandatory triggers for full re-examination and referral rather than simply “continuing feeds.” References:

- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;DIAGNOSTIC DECISION RULES AND RESOURCE-ADAPTED TEST INTERPRETATION;

Diagnostics in ##SAM## should function as a decision framework, not a shopping list. Normal or borderline results do not exclude severe disease, especially infection: clinical sepsis may exist without fever or striking inflammatory changes. Testing is mainly for syndromic localization, antimicrobial narrowing, and detection of dangerous electrolyte shifts during fluids and feed transition.

Phase the inpatient workup by urgency. Core tests when available are bedside glucose; hemoglobin (especially with severe palmar pallor); electrolytes including potassium and, during feed transition, phosphate; and a minimum infection screen (urinalysis with culture if feasible, blood culture when it will not delay antibiotics, and chest radiograph when respiratory signs or complications are suspected). Add malaria testing early in endemic settings or when fever/hypothermia has no clear focus.

If cultures, electrolytes, or imaging are unavailable, treat based on clinical syndrome pathways and monitor objectively (pulse and respiratory rate trends, temperature, mental state, urine output proxy, intake). During rehydration, rising respiratory rate or pulse from baseline should trigger stopping fluids and reassessment for overhydration rather than repeating boluses.

Use syndrome-based triggers to prioritize “essential” tests. Respiratory distress or hypoxia warrants chest radiograph when feasible but must not delay empiric antibiotics. Watery diarrhea with dehydration warrants electrolytes if available and structured reassessment. Dysentery requires empiric antibiotics without waiting for stool culture; culture is reserved for non-response or outbreaks. Suspected ##UTI## requires a properly collected urine sample for culture before antibiotics where possible. Failure to respond to verified feeding and antibiotics should trigger targeted evaluation for ##TB## and ##HIV## per national algorithms, recognizing that ##TST## can be falsely negative in severe malnutrition.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

- Indian Council of Medical Research, Standard Treatment Workflow: Paediatrics severe pneumonia — https://www.icmr.gov.in/icmrobject/uploads/STWs/1725952336_paediatrics_severe_pneumonia. pdf
- Indian Council of Medical Research, Standard Treatment Workflow: Paediatrics diarrhea — https://www.icmr.gov.in/icmrobject/uploads/STWs/1725952338_paediatrics_diarrhea.pdf - Indian Academy of Pediatrics, Standard Treatment Guidelines: Urinary tract infection in children — https://iapindia.org/pdf/STG-06-Urinary-Tract-Infection-in-Children.pdf - National Tuberculosis Elimination Programme, Paediatric TB management guideline — https://tbcindiawp.azurewebsites.net/wpcontent/uploads/2023/05/9534339438Paediatric_TB_Mangement_Gui deline_22082022_V1.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INPATIENT STABILIZATION AND EARLY MONITORING;MONITORING STOP RULES AND FAILURE TO RESPOND TIMEPOINTS;

Monitoring must be specified as frequencies with stop-rules, because deterioration and overhydration can evolve quickly in the first 24–72 hours of inpatient ##SAM## care. For any child receiving ##IV## fluids for shock, Indian ward thresholds operationalize safety: record pulse and respiratory rate at the start and at least every 5–10 minutes, and stop the infusion immediately if respiratory rate rises by 5/min or pulse rises by 15/min, then reassess perfusion and liver size. For oral or ##NG## rehydration without shock, reassess hydration and early overhydration every 30 minutes for the first 2 hours and then hourly; stop rehydration if new tachypnea, rising pulse, engorged neck veins, or periorbital puffiness develops.

For feeding, chart offered and taken volumes at every feed, vomiting, and stool frequency. Weigh daily at the same time and track oedema grade daily until resolution. Escalate to assisted feeding or ##NG## completion if intake remains below prescription despite support.

Use time-linked “failure to respond” triggers to force diagnostic reassessment. If appetite has not returned by day 4, reassess for missed infection, incorrect feed preparation or volume delivery, electrolyte derangements, or unrecognized comorbidities such as ##TB##, ##HIV##, urinary infection, parasitosis, or chronic disease. If oedema has not started to reduce by day 4 or persists to day 10, re-check the diagnosis (nutritional oedema versus nephrotic syndrome or heart failure), evaluate for dermatosis/skin sepsis, and review sodium exposure. After transition to free feeding, failure to gain ≥5 g/kg/day for three consecutive days should trigger a structured check for sharing, missed night feeds, malabsorption or persistent diarrhea, and chronic infection, with referral if local resources are limited.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fb mc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition

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https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;FLUID AND BLOOD MANAGEMENT IN SAM;
SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;FLUID AND BLOOD MANAGEMENT IN SAM;SHOCK AND DEHYDRATION REHYDRATION DECISION ALGORITHMS;
Teach fluid management in ##SAM## as “no ##IV## unless shock,” because overhydration is a leading iatrogenic cause of death and physical signs of dehydration can be misleading. Indian facility algorithms define shock in a ##SAM## child as lethargy or unconsciousness plus cold hands and either delayed capillary refill (>3 seconds) or a weak, fast pulse. If criteria are met, start oxygen, correct hypoglycemia and hypothermia, give antibiotics for presumed sepsis, then give a cautious fluid trial with intensive monitoring.

A pragmatic regimen is 15 mL/kg of an appropriate isotonic-with-glucose fluid over 1 hour (for example, Ringer lactate with 5% dextrose or half-normal saline with 5% glucose). Record pulse and respiratory rate at baseline and at least every 5–10 minutes, and stop the infusion if respiratory rate rises by 5/min or pulse rises by 15/min, then reassess perfusion and liver size. If there is no improvement after the first hour, do not continue repeated boluses; manage as septic shock and consider blood transfusion rather than further fluid loading.

For dehydration without shock, rehydrate orally or by ##NG## tube and deliver fluids slowly while continuing ##F-75##. An operational schedule used in Indian guidance is 5 mL/kg every 30 minutes for 2 hours, then 5–10 mL/kg/hour for up to 10 hours, with reassessment every 30 minutes initially and then hourly. Use ##ReSoMal## or a malnutrition-adapted approach with diluted low-osmolarity ##ORS## plus added potassium/glucose, but in suspected cholera or profuse watery diarrhea use standard low-osmolarity ##ORS## prepared normally (not diluted and not ##ReSoMal##). Replace ongoing losses between feeds (about 50 mL after each loose stool in <2 years; 100 mL if older) and reassess if new tachypnea, rising pulse, periorbital puffiness, or neck vein engorgement develops. References:
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fb_mc_with_sam.pdf
- World Health Organization, Dehydration in severely malnourished children (ELENA guidance) — https://www.who.int/tools/elena/interventions/dehydration-sam
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;FLUID AND BLOOD MANAGEMENT IN SAM;TRANSFUSION THRESHOLDS AND OVERLOAD PREVENTION IN SEVERE ANEMIA;

Severe anemia is common in hospitalized ##SAM## and must be managed as a high-risk intervention because cardiac reserve is reduced and fluid overload can precipitate heart failure. Indian training materials define transfusion triggers as hemoglobin <4 g/dL (or ##PCV## <12%), or respiratory distress when hemoglobin is 4–6 g/dL. Use 10 mL/kg of packed red cells or fresh whole blood infused slowly over about 3 hours, with minimal additional fluids and careful thermal protection during transfusion.

Treat transfusion as a monitored procedure, not a routine order. Check pulse, respiratory rate, perfusion, and liver size at baseline and repeatedly during the infusion. Worsening tachypnea, rising pulse, new hepatomegaly, crackles, or increasing work of breathing should prompt immediate slowing or stopping of transfusion and urgent reassessment for transfusionassociated circulatory overload.

Avoid confusing trainees about diuretics. The “no diuretic” rule applies to nutritional oedema in ##SAM##: do not give diuretics to reduce kwashiorkor oedema. The mention of furosemide in some Indian protocols is a separate indication related to transfusion overload risk in a child with severe anemia and low cardiac reserve. If a diuretic is used in this narrow context, document explicitly that it is for transfusion-associated volume overload prevention or treatment, not for nutritional oedema, and only use it with close monitoring. When shock is suspected and there is no response to the initial cautious fluid trial, Indian guidance favors early transfusion over repeated boluses, reflecting the high likelihood of sepsis and anemia in this phenotype.

References:

- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;READY-TO-PRESCRIBE ANTIBIOTIC REGIMENS WITH STEP-DOWN CRITERIA;

Empiric antibiotics in ##SAM## must be “ready to prescribe,” because invasive infection can present without fever and delays increase mortality. Align doses to one local standard (e.g., ##MoHFW##/##NHM## facility-based guidance) and document renal-safety stop rules, especially for gentamicin.

For uncomplicated outpatient ##SAM## (passes appetite test, no danger signs), use oral amoxicillin 15 mg/kg per dose every 8 hours for 5 days, unless local policy recommends an alternative. Reassess early if appetite remains poor or danger signs emerge.

For complicated inpatient ##SAM##, a common first-line regimen is:

- Ampicillin 50 mg/kg per dose by ##IV## or ##IM## every 6 hours plus gentamicin 7.5 mg/kg once daily (##IV## or ##IM##), typically for 7 days.

- Add cloxacillin 100 mg/kg/day in 4 divided doses when extensive skin lesions or abscess suggest staphylococcal infection.

Escalate at 48–72 hours if septic shock is present, clinical status is worsening, or there is no improvement despite verified feeds and electrolytes. Options include cefotaxime 150 mg/kg/day in 3 divided doses or ceftriaxone 100 mg/kg/day in 2 divided doses, with regimen choice guided by local antibiogram and suspected focus.

Embed renal and step-down rules in the order set. If the child is not passing urine (oliguria/anuria), withhold further gentamicin doses and reassess hydration, perfusion, and renal function before restarting. Step down to oral therapy once the child is clinically improving, tolerating enteral feeds, and can complete the course reliably; do not discharge a child with unresolved complications or ongoing nutritional oedema.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
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- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;ANTIBIOTIC STEWARDSHIP AND SYNDROME REASSESSMENT IN SAM;

In complicated ##SAM##, assume bacterial infection even when typical signs are absent. Antibiotic prescribing should be standardized and time-linked to reassessment, so that trainees neither undertreat silent bacteremia nor prolong broad-spectrum therapy unnecessarily.

Culture first when feasible, but never delay antibiotics in an unstable child. If blood culture or urine culture can be collected promptly, do so before the first dose; if not, start therapy and document that cultures were deferred due to clinical instability. Use early physiologic danger signs (hypothermia, recurrent hypoglycemia, persistent lethargy) as infection triggers even without fever.

Reassess at 48–72 hours and again at day 5. At each checkpoint, verify feed delivery and electrolyte supplementation, review vitals and appetite trajectory, and re-examine for missed foci (pneumonia, ##UTI##, skin sepsis, otitis, meningitis). If improving, narrow therapy to culture results or the most likely focus and step down from parenteral to oral antibiotics once the child is clinically stable, tolerating feeds, and reliable follow-up is arranged.

Document intended duration by syndrome to avoid default “7 days for all.” Use minimum durations from facility protocols (for example, suspected sepsis at least 7 days; ##UTI## 7–10 days; meningitis at least 14–21 days; deep infections such as osteomyelitis around 4 weeks). If there is no improvement by day 5, treat this as a mandatory trigger to reassess for resistant organisms, wrong diagnosis, or occult comorbidity such as ##TB## or ##HIV##, and escalate care or refer as needed.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Council of Medical Research, Standard Treatment Workflow: Paediatrics severe pneumonia — https://www.icmr.gov.in/icmrobject/uploads/STWs/1725952336_paediatrics_severe_pneumonia. pdf
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Urinary tract infection in children — https://iapindia.org/pdf/STG-06-Urinary-Tract-Infection-in-Children.pdf
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;PNEUMONIA WITH HYPOXIA MANAGEMENT AND ANTIBIOTIC STEP-DOWN;

In ##SAM##, treat pneumonia aggressively because hypoxia can develop rapidly and infection signs may be muted. Suspect severe pneumonia when cough or difficult breathing is accompanied by chest indrawing, cyanosis, grunting, or low oxygen saturation (commonly <90%, or <94% when pneumonia is present and pulse oximetry is available).

Immediate actions: give oxygen, maintain warmth, check glucose and treat hypoglycemia, and start feeds as soon as the child is stable enough to swallow safely. Do not delay antibiotics for chest radiography if the child is clinically unstable.

Empiric antibiotics for admitted/complicated ##SAM## should align to one local standard. A common first-line regimen is injectable ampicillin plus gentamicin. Add anti-staphylococcal coverage (for example cloxacillin) when staphylococcal pneumonia is suspected, including postmeasles pneumonia or extensive infected skin lesions.

Investigations that change management: obtain a blood culture when feasible (preferably before antibiotics) in very ill children, and use chest radiograph to evaluate complications such as

effusion, collapse, or staphylococcal pneumatocele. Electrolytes are useful when there is dehydration, shock, or feed transition risk.

Step-down rules: switch to an oral agent when respiratory distress is resolving, oxygen requirement has stopped or is minimal, vital signs are stable, and the child is accepting oral feeds. Complete a total 5–7 days for uncomplicated severe pneumonia; use longer total courses when staphylococcal pneumonia or complications are suspected, and reassess at 48–72 hours if there is no improvement. References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- Indian Council of Medical Research, Standard Treatment Workflow: Paediatrics severe pneumonia — https://www.icmr.gov.in/icmrobject/uploads/STWs/1725952336_paediatrics_severe_pneumonia. pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;SUSPECTED MENINGITIS INITIAL MANAGEMENT AND ANTIBIOTIC DURATION;
Consider meningitis in any ##SAM## child with coma, convulsions, bulging fontanelle, neck stiffness (often subtle), persistent irritability, or unexplained altered sensorium. Treat this as an emergency because delayed therapy has high mortality and clinical signs may be atypical.

Immediate actions: stabilize airway and breathing, give oxygen, correct hypoglycemia, prevent hypothermia, and control seizures using standard pediatric emergency protocols. Avoid routine fluid boluses unless shock criteria are met; overhydration can worsen outcome in severely malnourished children with reduced cardiac reserve.

Empiric antibiotics should follow facility guidance for complicated ##SAM## with suspected meningitis. Use a third-generation cephalosporin (cefotaxime or ceftriaxone) with an aminoglycoside (such as amikacin) as per local protocol, and document a minimum duration of 14–21 days. Modify therapy to culture and sensitivity results when available and reassess clinically at 48–72 hours.

Lumbar puncture and cultures: if the child is stable, obtain blood culture and cerebrospinal fluid before antibiotics. If the child is unstable or convulsing, prioritize immediate antibiotics and defer lumbar puncture until it is safe.

Monitoring priorities include serial mental status, seizure recurrence, temperature, glucose, and respiratory trends, because aspiration, hypoglycemia, and hypothermia commonly coexist. Start

##F-75## as soon as the airway is protected, use ##NG## feeding if oral intake is unsafe, and refer early for intensive care support if there is persistent coma, refractory seizures, or progressive respiratory failure.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;DYSENTERY MANAGEMENT WITH SECOND-LINE ESCALATION;

Treat any visible blood in stool as dysentery until proven otherwise. Severe malnutrition increases risk of invasive bacterial disease and complications, and dehydration assessment can be unreliable; therefore management must combine prompt antibiotics with cautious, closely monitored rehydration and continued feeding.

Empiric antibiotics: start oral ciprofloxacin promptly. A widely used regimen is ciprofloxacin 15 mg/kg per dose twice daily for 3 days (maximum per local protocol). If the child is very ill, is unable to tolerate oral therapy, or has received a fluoroquinolone recently, use parenteral ceftriaxone (commonly 100 mg/kg/day once daily or in divided doses) for about 5 days, then step down to oral therapy when improving.

Fluids and feeds: continue breastfeeding and stabilization feeds (##F-75##) rather than stopping feeding. Replace losses slowly using malnutrition-appropriate oral/##NG## rehydration (##ReSoMal## or diluted low-osmolar ##ORS## with added potassium/glucose per protocol), with reassessment every 30 minutes initially, then hourly. Stop or slow fluids if respiratory rate or pulse rise from baseline or edema worsens.

Escalation at 48 hours: if there is no clinical improvement, reassess adherence and dosing, consider resistant shigellosis, and look for alternative diagnoses (amoebiasis, persistent enteropathy, ##Celiac## disease) or occult sepsis. Send stool culture when available, review local resistance patterns, and refer to a higher-level facility if dehydration, shock, or systemic toxicity develops.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

- Indian Council of Medical Research, Standard Treatment Workflow: Paediatrics diarrhea — https://www.icmr.gov.in/icmrobject/uploads/STWs/1725952338_paediatrics_diarrhea.pdf
- World Health Organization, Dehydration in severely malnourished children (ELENA guidance) — https://www.who.int/tools/elena/interventions/dehydration-sam
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;URINARY TRACT INFECTION CULTURE-FIRST MANAGEMENT IN SAM;
In ##SAM##, consider ##UTI## in fever without a focus, persistent hypothermia, vomiting, poor weight gain, or failure to improve on empiric antibiotics. Because clinical signs can be muted, a culture-first approach prevents both missed infection and unnecessary broad-spectrum escalation.

Obtain urine for urinalysis and culture before antibiotics whenever feasible. Use a properly collected specimen: suprapubic aspiration or catheterization in non-toilet-trained children, and midstream clean-catch in toilet-trained children. Do not use bag urine for culture interpretation in infants and young children.
Start empiric antibiotics guided by local antibiogram and illness severity. For stable children who can take oral feeds, use an oral cephalosporin (such as cefixime) or amoxicillin/clavulanate per protocol. For toxic children, vomiting, or suspected pyelonephritis, start parenteral therapy (such as cefotaxime or ceftriaxone) and ensure concurrent stabilization feeding; step down to oral once afebrile, improving, and accepting feeds.

Treat for 7–10 days in ##SAM## unless culture results mandate otherwise. De-escalate to the narrowest effective agent once sensitivities are available. If an aminoglycoside is used, embed renal-safety rules (withhold further doses in oliguria/anuria and reassess). Reassess for urinary obstruction, renal anomaly, or an alternative diagnosis if there is no improvement within 48–72 hours, and ensure follow-up planning for recurrent ##UTI## risk after discharge.
References:
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Urinary tract infection in children — https://iapindia.org/pdf/STG-06-Urinary-Tract-Infection-in-Children.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;COMPLICATED MEASLES MANAGEMENT AND VITAMIN A DOSING;

Malnourished children have higher risk of severe or complicated measles, including pneumonia, diarrhea with dehydration, and corneal injury. Manage measles in ##SAM## as a high-risk syndrome where bacterial superinfection can evolve while classic fever patterns are blunted.

Immediate care priorities are airway and breathing support, prevention of hypoglycemia and hypothermia, and continuation of therapeutic feeding. Treat dehydration cautiously using malnutrition-appropriate oral/##NG## rehydration and frequent reassessment; avoid routine ##IV## boluses unless shock criteria are met.

Antibiotics: do not treat measles as “viral only” in a complicated ##SAM## inpatient. If pneumonia, worsening respiratory distress, otitis, skin infection, or sepsis is suspected, start empiric antibiotics per complicated ##SAM## protocols. In post-measles pneumonia or extensive skin infection, ensure anti-staphylococcal coverage because staphylococcal complications are common in this context.

Vitamin A is part of complication prevention and corneal protection. Give age-banded high-dose vitamin A when measles is suspected or confirmed, or when there are eye signs of deficiency (50,000 ##IU## if <6 months; 100,000 ##IU## if 6–12 months; 200,000 ##IU## if >12 months), following local guidance on repeat dosing if a high dose was given in the previous month.

Infection control (isolation, hand hygiene) and catch-up immunization planning are discharge tasks: document measles vaccination status and schedule follow-up doses once the child is clinically stable.

References:

- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;MALARIA TESTING AND ARTESUNATE-BASED TREATMENT IN SAM;

In malaria-endemic settings, test early in a ##SAM## child with fever, hypothermia, altered sensorium, anemia, or no clear infection focus. Include smear or rapid diagnostic test in the triage lab set because malaria can mimic sepsis and coexist with bacterial infection.

Uncomplicated falciparum malaria: treat using artemisinin-based combination therapy per Indian dosing guidance (for example oral artesunate for 3 days with an appropriate partner drug, or artemether–lumefantrine in a 6-dose regimen over 3 days). Ensure the child can tolerate oral intake; if vomiting or poor intake persists, manage as severe disease.

Recognize severe malaria early: impaired consciousness, repeated convulsions, respiratory distress, shock, severe anemia, hypoglycemia, jaundice, or hemoglobinuria should trigger inpatient parenteral therapy and close monitoring.

Severe malaria is an inpatient emergency in ##SAM##. Use parenteral artesunate 2.4 mg/kg ##IV## at 0, 12, and 24 hours and then once daily until the child can swallow, then complete therapy with a full oral regimen as per protocol. Provide supportive care with oxygen if needed, frequent glucose monitoring, temperature control, and early feeding once stable.

Fluid strategy must reflect malnutrition physiology. Avoid aggressive bolus resuscitation unless shock criteria are met, because overhydration can precipitate heart failure. Manage severe anemia using transfusion criteria rather than “early iron.” Reassess at 48–72 hours for response and consider additional infection foci when fever or hypothermia persists despite correct antimalarial therapy.

References:

- Indian Academy of Pediatrics, Standard Treatment Guidelines: Malaria — https://iapindia.org/pdf/Ch-095-Malaria.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
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SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;TUBERCULOSIS SCREENING AND NTEP LINKAGE IN NONRESPONDERS;

Non-response in ##SAM## wards should trigger a structured search for ##TB## rather than repeated feed escalation alone. Suspect ##TB## when there is known contact, chronic cough, persistent fever, lymphadenopathy, weight loss, or failure to gain weight despite verified therapeutic feeding and control of acute infection.

Interpret tests with malnutrition physiology in mind. Severe malnutrition can cause falsenegative ##TST## due to secondary immunodeficiency, so a negative skin test must not close the case in a high-risk, non-responding child. Use a syndrome-based approach: review chest symptoms and exam, obtain chest radiograph when available, evaluate for extrapulmonary disease, and consider concurrent ##HIV## testing per national algorithms.

Specimen and program linkage should be explicitly taught. When indicated, collect microbiological specimens using gastric aspirate/lavage or induced sputum per pediatric protocols, and send to the linked diagnostic facility through the established ##NTEP## network. Coordinate with the facility ##NRC## team to ensure the child is logged as a presumptive case and that results are actively tracked.

Do not discharge a non-responding child without a documented ##TB## plan. Record screening and referral actions in the ward record so that evaluation is not lost during transfer, and ensure the caregiver knows where and when to return for results.

Co-management: if ##TB## treatment is started, give pyridoxine (vitamin B6) prophylaxis with isoniazid-containing regimens (commonly 10 mg/day) because malnutrition increases neuropathy risk. Anticipate slower weight recovery and ensure close follow-up between nutrition services and ##NTEP## after discharge.
References:
- National Tuberculosis Elimination Programme, Paediatric TB management guideline — https://tbcindia-wp.azurewebsites.net/wp-content/uploads/2023/05/9534339438Paediatric_TB_Mangement_Guideline_22082022_V1.pdf - Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/
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SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;EXTENSIVE SKIN INFECTION MANAGEMENT AND ANTISTAPHYLOCOCCAL COVERAGE;
Extensive skin breakdown in oedematous ##SAM## is not only a nursing problem; it is a highrisk infection focus that can seed bacteremia while the child remains afebrile. Treat weeping kwashiorkor dermatosis, ulceration, spreading cellulitis, or abscess as systemic infection risk and reassess antibiotic coverage early.

Antibiotic approach: for admitted/complicated ##SAM## start standard parenteral coverage (for example ampicillin plus gentamicin). Add anti-staphylococcal coverage (for example cloxacillin) when lesions are extensive, purulent, rapidly spreading, or associated with post-measles pneumonia, because staphylococcal disease is common in these contexts. Escalate at 48–72 hours if there is deterioration or no improvement, guided by local antibiogram and culture where available.

Wound-care principles should be explicitly prescribed. Handle skin gently, avoid adhesive tapes on fragile areas, keep bedding soft and dry, and reduce pressure injury with frequent repositioning. Clean lesions with gentle soap-and-water or protocol-appropriate soaks, then apply a barrier dressing (petrolatum or zinc-based) to reduce fluid loss and bacterial entry. Avoid caustic topical antiseptics that delay healing.

Monitor for dehydration and hypothermia because open lesions increase insensible losses. Ensure adequate analgesia to support intake and sleep, treat scabies or fungal involvement when suspected, and document daily skin assessment as part of the infection reassessment checkpoint in complicated ##SAM##.
References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;POTASSIUM MAGNESIUM AND SODIUM RESTRICTION IN EARLY SAM CARE;

Electrolyte correction in ##SAM## is an early lifesaving intervention, not an optional add-on, because total-body potassium and magnesium deficits contribute to weakness, arrhythmia risk, and the physiology of nutritional oedema even when serum values appear “normal.” Start repletion on day 0, restrict sodium, and avoid diuretics for nutritional oedema.

A practical ward target used in standard protocols is potassium 3–4 mEq/kg/day for at least 2 weeks. Give potassium in divided doses with feeds (or as a syrup diluted into therapeutic milk) and review dosing if renal failure or anuria is suspected. Magnesium repletion can be started with 50% magnesium sulfate 0.3 mL/kg by ##IM## once on day 1 (maximum 2 mL), followed by oral magnesium 0.4–0.6 mmol/kg/day for about 2 weeks. If oral preparations are unavailable, small measured volumes of magnesium sulfate can be mixed into feeds per facility protocol.

Hard-code sodium restriction. Use “no added salt” feeds, avoid high-sodium family foods, and do not use standard high-sodium ##IV## fluids except as part of a cautious shock protocol. Interpret early weight change in oedematous ##SAM## carefully: initial weight loss may represent oedema resolution while electrolyte deficits are being corrected.

Monitor for overhydration and cardiac intolerance during rehydration and feed transitions (rising respiratory rate, increasing pulse, new hepatomegaly, or periorbital puffiness). Where laboratory monitoring exists, include phosphate during transition feeding, because hypophosphatemia can signal refeeding syndrome and should prompt slowed feed advancement plus urgent electrolyte correction.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_f bmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition
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- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;MICRONUTRIENT SUPPLEMENTATION SCHEDULE IN INPATIENT SAM CARE;

Micronutrient therapy in ##SAM## must be prescribed with attention to what the child is actually receiving. If therapeutic foods are correctly prepared and meet specifications (##F75##, ##F-100##, or ##RUTF##), additional multivitamins and trace elements may be unnecessary and can cause duplication. When local milk-sugar-oil recipes or home foods are used, supplementation must be explicit, reliable, and documented.

Use a standard inpatient supplementation schedule aligned to Indian facility protocols during stabilization and early rehabilitation. Provide folic acid 5 mg on day 1 followed by 1 mg daily. Provide zinc (commonly 2 mg/kg/day) and copper (commonly 0.3 mg/kg/day) for at least 2 weeks, and give a daily multivitamin preparation that includes vitamin B complex, vitamin C, and vitamin D per local formulary.

Link supplementation to feeding delivery. Give oral micronutrients with feeds to reduce gastric intolerance, and ensure the nursing intake chart includes a medication checklist so that micronutrients are not omitted during night shifts. If vomiting or severe diarrhea limits oral delivery, prioritize stabilization feeding and infection management first, then reintroduce supplements once tolerance improves.

Separate “routine schedule” from “condition-triggered therapy.” Iron and high-dose vitamin A have specific timing and indication rules in ##SAM## and should be prescribed as dedicated orders rather than being buried inside a generic multivitamin instruction for safety. References:

Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf

World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/

Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;IRON INITIATION CRITERIA AND DOSING DURING REHABILITATION;

Iron is delayed in ##SAM## because early supplementation during stabilization can increase oxidative stress and may worsen outcomes when infection is uncontrolled. Teach iron as a

rehabilitation-phase intervention with explicit prerequisites rather than as part of the admission order set.

- Start elemental iron only after the child is clinically stable and clearly improving: appetite has returned, hypothermia and hypoglycemia have not recurred, edema is reducing or resolved, and acute infection is treated and improving. In many teaching protocols this corresponds to about day 7 of inpatient care, but local operational guidance may start earlier; if so, require documentation that the child is in the rehabilitation phase and that infection is controlled.
- Prescribe a practical dose of elemental iron around 3 mg/kg/day (per local formulary) and continue through rehabilitation with follow-up planning. Pair iron with ongoing folate and nutrition rehabilitation rather than escalating dose to “force” hemoglobin recovery.
- Use transfusion criteria for severe anemia instead of early iron. If hemoglobin is <4 g/dL (or if there is respiratory distress with hemoglobin 4–6 g/dL), manage with cautious blood transfusion and close monitoring per facility protocol. Treat persistent poor weight gain or failure of anemia to improve as a trigger to reassess for ongoing infection, chronic inflammation, parasitosis, or hemoglobinopathy, and verify that therapeutic feeding and micronutrient delivery are actually occurring.
- Do not add extra iron when the child is receiving correctly formulated ##RUTF## unless a separate clinical indication exists, because duplication can increase adverse effects and reduce adherence. References:
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;VITAMIN A SUPPLEMENTATION INDICATIONS IN SAM; Vitamin A management in ##SAM## should be indication-based and linked to eye findings and measles risk. Routine repeat megadosing can cause duplication when fortified therapeutic foods are used, while missed dosing in measles or corneal disease can cause blindness. - Assess at admission for conjunctival xerosis, Bitot spots, corneal clouding or ulceration, and a history of recent high-dose vitamin A. Treat any corneal involvement as an emergency requiring urgent ophthalmic care and vitamin A therapy.

- Give high-dose vitamin A on day 1 when there are eye signs of deficiency or when measles is suspected or confirmed. Use age bands commonly applied in pediatric protocols: 50,000 ##IU## if <6 months, 100,000 ##IU## if 6–12 months, and 200,000 ##IU## if >12 months. Follow local/national guidance for a repeat dose (often after 24

hours and again after 2 weeks in frank xerophthalmia), but avoid repeat dosing if a high dose was given in the preceding month unless there is a specific indication.

- In edematous ##SAM## with dermatosis, ensure vitamin A status is actively considered because skin and mucosal integrity are compromised. In complicated measles or pneumonia, do not delay vitamin A while waiting for laboratory confirmation.
- On discharge, document whether a high dose was given and communicate this to the outpatient team to prevent duplicate dosing. Distinguish vitamin A therapy from measles vaccination: vitamin A is a complication-prevention treatment, while measles vaccination is a separate preventive action once the child is clinically stable. References:
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFECTION MANAGEMENT AND METABOLIC CORRECTIONS;DEWORMING TIMING AND ANTIHELMINTHIC DOSES DURING REHABILITATION;

Deworming is a rehabilitation-phase intervention in ##SAM##, not a day-0 priority, because early inpatient care should focus on hypoglycemia, hypothermia, infection treatment, and careful feeding. Anthelmintics can worsen vomiting in an unstable child and do not address the immediate metabolic fragility that drives early deaths.

Plan deworming once the child is clinically stable, has transitioned to catch-up feeding, and can tolerate oral medicines reliably (commonly after the first 48–72 hours of stabilization or at discharge). Use age-appropriate dosing per Indian protocols: albendazole 200 mg as a single dose at 12–23 months and 400 mg as a single dose at ≥24 months. Where albendazole is unavailable or local policy prefers, mebendazole 100 mg twice daily for 3 days can be used in children ≥24 months. Avoid deworming in children <12 months unless directed by a specific program policy. If persistent diarrhea is present, stool examination for giardiasis or other pathogens should not be replaced by empiric deworming.

Link deworming to the outpatient follow-up workflow. Document the dose and date given, and pair the drug with household hygiene counseling (handwashing, safe disposal of child feces, footwear where feasible) to reduce reinfection. Counsel caregivers that mild transient abdominal discomfort can occur, but treat persistent vomiting, abdominal distension, rash, or worsening anorexia after dosing as a reason for reassessment rather than repeating the drug. In settings with high anemia burden, do not assume helminths are the only cause; continue evaluation for dietary deficiency, infection, and severe anemia requiring transfusion per ##SAM## protocols.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;WEIGHT-BASED F-75 PRESCRIPTION AND FEED VOLUME TARGETS;
Stabilization feeding in ##SAM## is a prescription, not “nutrition advice.” The aim in the first days is metabolic stabilization and prevention of hypoglycemia, not rapid catch-up growth. Use low-sodium, low-protein ##F-75## (or equivalent starter diet) given day and night in small, frequent feeds.

Write orders in executable bedside terms: “##F-75## ___ mL per feed every ___ hours; daily target ___ mL; minimum acceptable intake (80% of target) ___ mL/day; if intake <80% initiate ##NG## top-up.” Indian facility charts commonly target about 130 mL/kg/day total fluid (reduce to about 100 mL/kg/day in severe (+++) edema), starting 2-hourly feeds for at least the first day and moving to 3-hourly or 4-hourly only when vomiting and diarrhea are controlled and feeds are finished.

Feeding progression rules should be explicit. Do not increase volumes because the child is “hungry” on day 1; instead, maintain prescribed volumes and frequency until temperature is stable, hypoglycemia has not recurred, and appetite is improving. If the child vomits, reduce volume per feed and increase frequency rather than skipping feeds. Continue breastfeeding on demand alongside therapeutic feeds.

Document each feed offered and taken, with leftover volume recorded. If two consecutive feeds are ≤80% of the prescription despite assisted feeding, treat this as a trigger for ##NG## completion and reassessment for sepsis, electrolyte derangement, or intestinal pathology. Do not administer ##RUTF## through an ##NG## tube; use liquid therapeutic feeds for tube delivery.
References:
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;F-75 AND F-100 PREPARATION WITH ELECTROLYTE FORTIFICATION;
Therapeutic feeding safety depends on correct preparation and mineral composition, not only calories. When premixed ##F-75##/##F-100## is unavailable and local recipes are used, trainees must be taught that electrolytes and micronutrients must still be delivered reliably.

Facility-style recipes commonly used in India include a starter diet (##F-75## equivalent) prepared from milk, sugar, oil, and water, and a catch-up diet (##F-100## equivalent) prepared from higher milk content with added sugar and oil. Preparation should be done with clean utensils, boiled or safe water, and batch labeling with time of preparation; discard feeds kept beyond the protocol window to reduce bacterial contamination.

Electrolyte fortification: severe malnutrition requires added potassium and magnesium and restricted sodium. Where an electrolyte/mineral solution is available, mix it into therapeutic milk as directed so that each feed carries routine supplementation. Where it is not available, prescribe potassium and magnesium separately (and document dosing), because waiting for lab confirmation risks arrhythmia and feeding failure.

Practical ward checks that prevent errors include double-checking volume calculations against weight, verifying that oil and sugar quantities match the recipe (common under-measurement causes poor weight gain), ensuring feeds are warmed before administration, and confirming that night feeds are delivered. If diarrhea worsens after preparation changes, treat it as a hygiene and mixing-process problem until proven otherwise and retrain the caregiver before discharge.
References:
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Integrated management of acute malnutrition, 2010 — https://platform.who.int/docs/default-source/mca-documents/policy-documents/guideline/ERICH-38-01-GUIDELINE-2010-eng-Integrated-Management-Acute-Malnutrition.pdf - World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;FEEDING INTOLERANCE TROUBLESHOOTING AND NURSING DOCUMENTATION;
Feeding intolerance in ##SAM## should trigger protocol adjustments and diagnostic reassessment, not cessation of feeds. The ward goal is to maintain frequent small feeds, prevent fasting, and identify the correct reason for poor tolerance (incorrect volumes, cold feeds, sepsis, electrolyte derangement, or evolving heart failure).

Vomiting in the first 48 hours: reduce volume per feed, return to 2-hourly feeds, and ensure feeds are warm. Recheck glucose and temperature and search for infection or obstruction if vomiting persists. Do not “push through” by increasing volume or switching early to catch-up feeding.

Persistent watery diarrhea during stabilization: continue breastfeeding and ##F-75##; replace losses between feeds using malnutrition-appropriate oral/##NG## rehydration and frequent reassessment. If dehydration is suspected, treat slowly and monitor for overhydration signals (rising respiratory rate and pulse, periorbital puffiness, worsening edema).

Abdominal distension, new tachypnea, or rising pulse during feeding should be treated as possible fluid overload or early heart failure. Reduce feeds and stop additional fluids, reassess liver size and perfusion, and review sodium exposures and feed preparation.

Nursing documentation is a safety intervention. Each shift should record: temperature, pulse and respiratory rate trends, feed offered and taken at every feed, vomits and stools, urine output proxy, edema grade, and any overhydration signs. Intake <80% of prescription over 24 hours, recurrent hypoglycemia, or new respiratory distress are escalation triggers that must be acted on, not only charted.
References:
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Dehydration in severely malnourished children (ELENA guidance) — https://www.who.int/tools/elena/interventions/dehydration-sam
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;NASOGASTRIC FEEDING INDICATIONS PLACEMENT CONFIRMATION AND ASPIRATION PREVENTION;
In inpatient ##SAM##, ##NG## feeding is a safety tool to achieve minimum intake when appetite or endurance is poor. Use it early when oral intake is unreliable, rather than allowing repeated missed feeds and recurrent hypoglycemia.

Indications include inability to finish prescribed ##F-75## volumes, intake persistently <80% of target despite assisted feeding, weak suck or unsafe swallow, and the need to give glucose or feeds when the child cannot drink. Contraindications needing senior decision-making include depressed consciousness with poor airway protection, suspected upper gastrointestinal obstruction, severe facial trauma, and uncontrolled vomiting.

Placement should be confirmed before first use. Measure insertion length using a standard nose-to-earlobe-to-mid-umbilicus method and mark external length. Use gastric aspirate pH testing when available; feeding is generally considered safe when aspirate pH is ≤5.5. If aspirate cannot be obtained or pH is higher, do not use the tube and obtain radiographic confirmation or re-site the tube per pediatric safety guidance. Re-check position after vomiting, violent retching, or suspected displacement.

Aspiration prevention is practical bedside behavior: keep the head elevated during feeding, avoid supine feeding, slow bolus delivery, and pause feeds during significant respiratory distress. Plan weaning when the child reliably takes most feeds orally (commonly ≥80% of target over 24 hours) without vomiting and with stable respiratory status.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Children's Hospitals Solutions for Patient Safety, Blind pediatric NG tube placements alert — https://www.childrenshospitals.org/content/quality/alerts/blind-pediatric-ng-tubeplacements - Buckinghamshire Healthcare NHS Trust, Nasogastric tube feeding at home guideline — https://www.buckshealthcare.nhs.uk/wp-content/uploads/2025/02/NasogastricTube-Feeding-atHome.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;TRANSITION TO F-100 CATCH-UP GROWTH AND REFEEDING SAFETY;

Transition from stabilization feeding to rehabilitation feeding is a monitored intervention because overly rapid escalation can precipitate heart failure in ##SAM##. Begin rehabilitation only when appetite has returned, temperature is stable, there has been no further hypoglycemia, and oedema is reducing or resolved.

Use a structured transition from ##F-75## to catch-up feeding with ##F-100##. A practical schedule used in Indian facility guidance is to replace ##F-75## with an equal volume of ##F100## for about 2 days, then increase each feed gradually until small amounts remain uneaten (a bedside sign of ad lib tolerance). Maintain frequent feeding (often 8 feeds over 24 hours) and monitor for tachypnea, rising pulse, hepatomegaly, or recurrent oedema as warning signs of cardiac intolerance.

During rehabilitation, monitor catch-up growth as an outcome measure and a safety signal. Daily weight gain after starting catch-up feeding is commonly interpreted as good at ≥10 g/kg/day, moderate at 5 to <10 g/kg/day, and poor at <5 g/kg/day. Interpret early weight loss in oedematous ##SAM## cautiously because it may reflect oedema resolution. If gain is poor despite verified feed delivery, reassess for sharing or missed night feeds, persistent diarrhea or malabsorption, and chronic infection such as ##TB##.

Recognize and prevent refeeding syndrome. Rapid nutritional rehabilitation can trigger intracellular shifts of phosphate, potassium, and magnesium, causing arrhythmias, weakness, and respiratory failure. Prevention is adherence to phased feeding (##F-75## then ##F100##/##RUTF##), early potassium and magnesium repletion, and monitoring electrolytes including phosphate when available. If suspected, slow feed advancement and correct phosphate/potassium/magnesium urgently while monitoring cardiorespiratory status.
References:
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fb mc_with_sam.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;REFEEDING SYNDROME RECOGNITION THIAMINE AND PHOSPHATE REPLACEMENT;
Refeeding syndrome risk is predictable in ##SAM## because potassium, magnesium, and phosphate stores are depleted and carbohydrate-driven insulin release after feeding shifts these electrolytes intracellularly. Clinical deterioration can occur within 2–5 days of advancing feeds, even when calories appear “appropriate.”
Suspect refeeding syndrome when new tachypnea or hypoxia develops without another explanation, arrhythmia or sudden tachycardia occurs, neuromuscular weakness or lethargy worsens, seizures appear, or edema increases during feed advancement. Where labs are available, falling phosphate, potassium, and magnesium are key markers.

Immediate response: do not stop feeding completely unless airway protection is compromised. Reduce caloric advancement (temporarily revert toward stabilization volumes), increase feed frequency, and correct electrolytes proactively. Continue routine potassium (about 3–4 mmol/kg/day) and magnesium (about 0.4–0.6 mmol/kg/day) supplementation even without lab confirmation, and add thiamine before further escalation where feasible. Treat concurrent infection aggressively, because sepsis and refeeding often coexist.

Phosphate replacement is indicated when hypophosphatemia is moderate-to-severe or symptoms are present. Use enteral phosphate when gut function and aspiration risk allow, with divided dosing per pediatric monographs. Severe or symptomatic hypophosphatemia requires ##IV## phosphate with cardiac and calcium monitoring and dose reduction in renal dysfunction;

if phosphate products or testing are unavailable, slow feeding advancement and arrange urgent referral for electrolyte correction.

References:

- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- StatPearls, Refeeding Syndrome (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK493172/
- Children's Hospital Los Angeles, Refeeding guideline (GROW 2018) — https://www.chla.org/sites/default/files/atoms/files/Refeeding%20GROW%202018.pdf
- Leeds Formulary, Paediatric monograph: Oral phosphate — https://www.leedsformulary.nhs.uk/docs/Paediatric%20Monograph%20-%20Oral%20Phosphate.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;SENSORY STIMULATION AND EMOTIONAL SUPPORT DURING REHABILITATION;

Rehabilitation in ##SAM## includes developmental and psychosocial care, because apathy, irritability, and caregiver distress can reduce intake, worsen defaulting, and delay neurodevelopmental recovery even when therapeutic feeding is technically correct. Treat sensory stimulation and emotional support as a prescribed intervention that begins once the child is clinically stable.

Create a structured daily routine on the ward: encourage the caregiver to stay with the child, support responsive feeding (calm, unhurried feeding with encouragement but no force), and schedule short play sessions (for example 15–30 minutes, at least twice daily) using ageappropriate, low-cost toys and interaction (talking, singing, simple games). Allow safe mobility and sunshine exposure when stable, and use group play where feasible to reduce isolation and improve caregiver skill transfer. Document engagement and alertness alongside intake and weight gain; improving playfulness is often an early marker of recovery.

Identify barriers that predict relapse. Screen for maternal depression, household food insecurity, domestic violence, and disability-related feeding problems that require tailored counseling or referral. For children with prolonged anorexia, vomiting, or suspected oral pain, assess for thrush, dental problems, and painful skin lesions, and treat pain to restore intake. Before discharge or step-down to outpatient ##CMAM##, provide a simple home stimulation plan and reinforce that the child's appetite and interaction should continue to improve; deterioration in activity is a danger sign even if weight has not yet fallen.

References:

- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;THERAPEUTIC FEEDING AND CATCH-UP GROWTH;RUTF SPECIFICATIONS CAREGIVER COUNSELING AND SACHET DOSING;
##RUTF## is an energy-dense, micronutrient-fortified paste used mainly for outpatient treatment of uncomplicated ##SAM## in children >6 months. It is a monitored medical nutrition therapy, not a general complementary food. Do not administer ##RUTF## via ##NG## tube; inpatient ##NG## feeding should use liquid therapeutic formulas.
Counsel caregivers on safe use: 1 sachet provides 500 kcal; divide the prescribed daily sachets into multiple small feeds; offer clean water after feeds; continue breastfeeding on demand; do not share within the household; and discard an opened sachet after 24 hours. Give clear “return today” danger signs (new fever, fast breathing, convulsions, lethargy, repeated vomiting preventing intake, new/worsening oedema, or inability to take the prescribed ##RUTF##) and schedule at least weekly follow-up for re-measurement and resupply.

Translate energy targets into sachets. The 2023 ##WHO## dosing framework allows either 150–185 kcal/kg/day until full recovery, or 150–185 kcal/kg/day until the child is no longer severely wasted and has no nutritional oedema, then 100–130 kcal/kg/day until recovery. A practical starting prescription targeting 150–185 kcal/kg/day (rounded to the nearest halfsachet) is:
4.0–5.0 kg: 1.5 sachets/day 5.5–6.7 kg:
2.0 sachets/day
6.8–8.3 kg: 2.5 sachets/day
8.4–10.0 kg: 3.0 sachets/day
10.1–11.7 kg: 3.5 sachets/day
11.8–13.3 kg: 4.0 sachets/day
Adjust only within a protocol: discrete sachets make exact kcal/kg matching impossible at some weights, and dose reduction is unsafe when household food security or follow-up is unreliable.
References:

- UNICEF Supply Division, Ready-to-use therapeutic food specification sheet (1 sachet = 500 kcal) — https://www.unicef.org/supply/media/18021/file/S0000240-RUTF-specification.pdf - World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- childwasting.org, Briefing note on simplified approaches to treatment of wasting — https://www.childwasting.org/media/1501/file/Briefing_note_Simplified_Approaches.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;
SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;THERAPEUTIC FOOD AVAILABILITY STOCK-OUT ADAPTATIONS AND VISIT WORKFLOW;

Therapeutic food availability in India is heterogeneous and should be stated explicitly because it shapes what “outpatient ##SAM## treatment” looks like. National technical reviews note that community-based management depends on access to a therapeutic food, yet India has not had a single universally adopted product for ##SAM## across states. Consequently, districts may use ##RUTF##, locally produced equivalents, or state-designed supplements, while inpatient care commonly relies on locally prepared ##F-75## and ##F-100## recipes from ##MoHFW## guidance.
For clinicians, the safety-critical message is that protocol reliability matters more than brand: correct triage and disposition, cautious medical management (fluids, antibiotics, electrolytes), delivery of adequate energy/protein/micronutrients, and strict follow-up with escalation rules. When therapeutic foods are constrained or stock-outs occur, programs should predefine adaptation rules rather than improvising at the bedside. The 2023 ##WHO## framework describes structured dose-reduction only after a child transitions out of ##SAM## criteria and has no nutritional oedema, and only when complementary foods and follow-up are reliable.

A standardized outpatient ##CMAM## visit should function like a brief repeatable ward-round. At every visit: screen danger signs and new/worsening oedema for immediate referral; verify appetite and practical feeding feasibility; measure weight for dosing and track ##MUAC##/##WHZ## response when available; and assess predictable household risks (sharing, missed days, stock-outs, travel barriers, intercurrent illness). At entry, document a focused exam and a standardized appetite test with the therapeutic food planned for home use, initiate caregiver demonstration and hygiene counseling, and schedule follow-up at least weekly until recovery momentum is clear. References:

- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf - Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;OUTPATIENT ESCALATION DEFAULTING AND DISCORDANT ANTHROPOMETRY DECISIONS;
Outpatient programs must hard-code escalation criteria because deterioration can be fast and infection signs can be muted in ##SAM##. Refer immediately for inpatient stabilization if any new danger sign develops, the appetite test fails, oedema appears or worsens, repeated vomiting prevents intake, severe dehydration or shock is suspected, or the caregiver cannot administer feeds safely. Treat persistent anorexia as a warning signal: lack of appetite despite initial outpatient antibiotics should trigger reassessment for missed infection or incorrect feeding delivery rather than simply renewing the same plan.

Defaulting and sharing are predictable system risks, not moral failures. Large Indian program experience has shown high defaulting when travel is difficult or when community referral is weak; referrals by frontline workers such as ##ASHA## reduce default risk, highlighting that follow-up support and case tracing are clinical safety tools. Incorporate short adherence questions at every visit ("who else eats the therapeutic food," "any days missed," "any stockouts," "can you return next week") and treat problems as indications for intensified counseling and follow-up.
Manage discordant anthropometry explicitly. ##MUAC## and ##WHZ## do not identify identical risk groups, so enroll a child if either criterion meets ##SAM## threshold or if nutritional oedema is present. For discharge, avoid a single-marker decision: do not discharge solely because ##MUAC## has normalized if ##WHZ## remains <−3, and do not discharge on ##WHZ## improvement alone if ##MUAC## remains severely low. A safe rule is to document both measures at baseline and at planned discharge, require absence of nutritional oedema on repeated visits, and require sustained clinical improvement and weight gain trajectory alongside anthropometric recovery. References:

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/Statement-WHO-WFP-SCNandUNICEFonCommunit y-Based-Management-of-SAM.pdf
- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;OUTPATIENT REFERRAL AND ESCALATION CRITERIA IN CMAM;
Outpatient ##CMAM## is safe only when deterioration triggers rapid referral. Escalation criteria should be treated as patient-safety rules, because infection signs can be muted and weight change can be distorted by edema and intercurrent illness.

- Refer immediately for inpatient stabilization if any danger sign develops (lethargy, convulsions, hypothermia, hypoglycemia, respiratory distress), if the appetite test fails, if bilateral pitting edema appears or worsens, or if repeated vomiting prevents intake. Treat suspected severe dehydration or shock as an inpatient indication, not as an outpatient "##ORS## trial," because overhydration risk and diagnostic uncertainty are high in ##SAM##.
- Escalate when recovery momentum is absent. Lack of appetite despite correct outpatient antibiotics and verified feeding should trigger clinical reassessment for missed infection, incorrect therapeutic food delivery, or caregiver barriers, rather than automatic renewal of the same plan. Weight loss between visits, new edema, or failure to gain over two successive visits should be treated as high-risk non-response and prompt referral. - Practical "social" triggers are also medical triggers. If the caregiver cannot demonstrate safe, frequent feeding; cannot return reliably for at least weekly follow-up; reports stock-outs or sharing that cannot be

corrected; or has unsafe water/sanitation that makes home feeding unsafe, start inpatient initiation or supervised bridging rather than persisting with outpatient care. References:

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/Statement- WHO-WFP-SCN-and-UNICEF-onCommunity-Based-Management-of-SAM.pdf
- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-AcuteMalnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;MANAGING DEFAULTING AND ADHERENCE FAILURES IN CMAM;

Defaulting in ##CMAM## is a clinical risk signal, not only a reporting outcome. Missed visits predict missed doses of therapeutic food and delayed recognition of infection or edema relapse, so tracing and adherence support should be built into the care plan.

- At each visit, ask short, concrete adherence questions: any days missed; who else ate the therapeutic food; any vomiting or refusal; any stock-outs; and ability to return next week. Treat identified barriers as indications for intensified counseling, additional home visits, or supervised feeding rather than blaming caregivers.
- A practical tracing algorithm for Indian programs is: after one missed visit, trigger same-week outreach by ##ASHA##/##AWW## (home visit or phone contact) and rebook within 48–72 hours; after two missed visits, escalate to supervisor/##ANM## review and assess for illness or social barriers; if danger signs, weight loss, or edema are suspected, refer to facility immediately. Use community platforms such as

##VHSND## days or automated messaging where available to reinforce return visits. Document tracing attempts and outcomes to reduce “silent defaults.”

- Stock-outs require a pre-defined playbook. Use routine stock reconciliation and early warning systems, arrange redistribution between centers when feasible, and provide protocol-approved temporary bridging only when follow-up is reliable. During supply interruption, treat failure to gain weight or any clinical deterioration as an indication for urgent facility referral rather than “waiting for the next consignment.” References:
- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ -

The Sphere Project, Sphere Handbook 2011 — https://www.ndsu.edu/sites/default/files/fileadmin/emgt/The_Sphere_Project_Handbook_2011.pdf
- WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management
of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/StatementWHO-WFP-SCN-and-UNICEFonCommunity-Based-Management-of-SAM.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;DISCHARGE DECISIONS WHEN MUAC AND WHZ DISAGREE; ##MUAC## and ##WHZ## identify overlapping but non-identical high-risk groups, and discordant recovery is common after prolonged treatment. Discharge decisions must therefore be anchored to the admission criterion and repeated edema-free stability, not a single improved number.

- At enrollment, document which criterion triggered admission to ##CMAM## (##MUAC## <115 mm, ##WHZ##/##WLZ## <−3, and/or nutritional edema). Use the same criterion as the primary recovery anchor. A safe ##WHO## recovery standard is: ##MUAC## ≥125 mm and no bilateral pitting edema for at least two consecutive visits, or ##WHZ##/##WLZ## ≥−2 and no edema for at least two consecutive visits, depending on the program pathway.
- When indicators disagree, do not discharge solely because one normalizes. If ##WHZ## improves to ≥−2 but ##MUAC## remains <125 mm, continue therapeutic feeding and close follow-up because residual muscle wasting still carries substantial mortality risk. If ##MUAC## normalizes but ##WHZ## remains low, reassess measurement quality and continue treatment; look for chronic infection, malabsorption, disability-related feeding difficulty, or household sharing.
- Combine anthropometry with functional recovery: sustained appetite, activity, and caregiver competence in safe feeding. In India, where community screening practices may vary, documenting both ##MUAC## and weight-for-height during clinic visits reduces the risk that a child is discharged from one system while still meeting criteria in another.

References:

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf - NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf
- WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management
of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/StatementWHO-WFP-SCN-and-UNICEFonCommunity-Based-Management-of-SAM.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;OUTPATIENT CMAM IMPLEMENTATION IN INDIA;OUTPATIENT RECOVERY CRITERIA AND MAXIMUM DURATION WITH NON-RESPONSE REFERRAL;

Outpatient ##CMAM## programs must state recovery and exit criteria explicitly and label them as “program cure,” distinct from inpatient “transfer readiness.” For children treated as outpatients, the safest cure logic is anthropometric recovery plus a sustained oedema-free period, because oedema can relapse and weight can fluctuate with intercurrent illness.

- Use one primary recovery standard consistently. A widely used ##WHO## criterion for discharge from nutritional treatment is: ##MUAC## ≥125 mm with no bilateral pitting oedema for at least 2 weeks, or ##WHZ## ≥−2 with no bilateral pitting oedema for at least 2 weeks. Do not use percentage weight gain (e.g., “15% gain”) as a sole cure criterion, because it is biased by admission weight (including oedema) and does not ensure normalization of wasting risk. Document both ##MUAC## and weight at each visit for dosing and trend interpretation, even if only one indicator is used for admission.

- Define maximum length of stay and non-response triggers to prevent silent failure.

Indian state protocols vary, but many program designs use a maximum duration of about 12–16 weeks for outpatient ##SAM## treatment, with referral for medical evaluation when recovery does not occur within the defined window. Earlier referral should occur if there is weight gain <5 g/kg/day over several weeks despite verified adherence, persistent or worsening oedema, repeated appetite-test failure, or recurrent infections. Classifying a child as “non-responder” should trigger a structured search for missed infection, chronic diarrhea or malabsorption, ##TB##/##HIV##, disability-related feeding difficulty, or unsafe home delivery of therapeutic feeds. References:

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/Statement-WHOWFP-SCNandUNICEFonCommunity-Based-Management-of-SAM.pdf
- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf
- The Sphere Project, Sphere Handbook (minimum standards in humanitarian response)

— https://www.ndsu.edu/sites/default/files/fileadmin/emgt/The_Sphere_Project_Handbook_2011.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DISCHARGE FOLLOW-UP AND PROGRAM QUALITY;

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DISCHARGE FOLLOW- UP AND PROGRAM QUALITY;INPATIENT DISCHARGE READINESS AND STEP-DOWN HANDOVER CRITERIA;

Discharge from inpatient ##SAM## care is a handover between risk environments, not an endpoint. Teach “discharge readiness” as a clinical stability threshold (safe to continue rehabilitation outside the ward), and separate it from outpatient program “cure” criteria, which require anthropometric recovery and an oedema-free duration.

Use functional inpatient discharge readiness criteria consistent with Indian facility protocols: complications treated and clinically resolving; child alert with good appetite and able to take prescribed feeds by mouth; no hypoglycemia or hypothermia for at least 24 hours; dehydration corrected without ongoing shock; and nutritional oedema clearly reducing or resolved. A practical feeding readiness marker is sustained intake delivering roughly 120–130 kcal/kg/day with demonstrated weight gain ≥5 g/kg/day for 3 consecutive days on oral feeds. Complete parenteral antibiotics before discharge unless there is a supervised step-down plan. Make the discharge order explicit. Document the current therapeutic feeding prescription (local catchup diet or ##RUTF## sachet dose), provide written “return today” danger signs (poor intake, repeated vomiting, fever, fast breathing, convulsions, lethargy, new/worsening oedema), and schedule the first follow-up within 7 days. Link families to the nearest outpatient nutrition service and document referral pathways back to inpatient stabilization if appetite fails or complications recur. Reinforce that early weight loss in oedematous ##SAM## can reflect oedema resolution; clinical status and oedema grade must guide escalation decisions. References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IEC-materials/PARTICIPANT-MANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DISCHARGE FOLLOW-UP AND PROGRAM QUALITY;DISCHARGE COUNSELING CHECKLIST AND OUTPATIENT PRESCRIPTION TEMPLATE;

Discharge planning in ##SAM## is a handover of risk, not an endpoint. A discharge counseling checklist and a structured outpatient prescription reduce relapse, defaulting, and unsafe home feeding.

Before discharge or step-down, require caregiver demonstration: correct preparation and frequency of the prescribed feed (local catch-up diet or ##RUTF## sachet schedule), safe storage and hygiene, and accurate administration of medicines and micronutrients. Confirm the caregiver can recognize danger signs that require same-day return (poor intake, repeated vomiting, fever or hypothermia, fast breathing, convulsions, lethargy, new or worsening edema) and knows the nearest referral facility.

Write the outpatient prescription as a complete plan rather than a product name. Specify the therapeutic food and daily dose (for ##RUTF##, number of sachets/day and how to divide them), continuation or completion dates for antibiotics if applicable, micronutrient schedule with iron timing rules, and the exact next visit date (ideally within 7 days). Include a tracing trigger: “if visit missed, contact family within 48–72 hours,” with the responsible cadre (##ASHA##/##AWW##/##ANM##) named according to local program design.

Document the discharge anthropometry and edema status and communicate whether high-dose vitamin A was given to avoid duplication. Where outpatient services are supply-constrained, counsel families on what to do if therapeutic food is temporarily unavailable and provide clear escalation instructions if weight gain stalls or the child becomes unwell.

References:

- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DISCHARGE FOLLOW- UP AND PROGRAM QUALITY;MEASLES VACCINATION TIMING AND ROUTINE IMMUNIZATION CATCH-UP;

Severe malnutrition is an immunization-risk state, so vaccination review should be a standard action during ##SAM## treatment rather than deferred indefinitely. Build immunization status checks into both inpatient admission and discharge workflows, and into outpatient ##CMAM## follow-up visits.

Measles prevention deserves explicit teaching because measles case fatality is higher in malnourished children. In standard hospital guidance, give a measles-containing vaccine to eligible children with ##SAM## who are ≥6 months and are unvaccinated or incompletely vaccinated, but delay vaccination if the child is in shock or has uncontrolled acute instability. If a dose is given before 9 months (an “early dose” for high-risk children), ensure the child still receives routine measles-containing doses according to the national schedule after 9 months, because early vaccination may not provide durable immunity.

Operationalize this as a discharge task: document the immunization card status, administer due vaccines once the child is clinically stable, and provide a clear plan for remaining doses (date and location). In outpatient care, reassess immunization status at each weekly visit; link families to routine immunization sessions and treat missed appointments as a follow-up failure risk similar to missed therapeutic feeding visits. Do not confuse vaccination timing with vitamin A megadosing: vitamin A should follow indication-based protocols (eye signs or suspected

measles) when therapeutic foods are used, while measles vaccination is a preventive action once the child is stable.

References:

- World Health Organization, Pocket book of hospital care for children: severe acute malnutrition chapter (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/childhealth/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/childhealth/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;DISCHARGE FOLLOW- UP AND PROGRAM QUALITY;CMAM PROGRAM PERFORMANCE INDICATORS AND SPHERE BENCHMARKS;

Program quality metrics in ##SAM## are patient-safety metrics, because poor cure rates and high defaulting often reflect missed illness, unreliable therapeutic food delivery, or delayed referral. Benchmarks must be labeled by setting: inpatient stabilization units (e.g., ##NRC##style facilities) and outpatient ##CMAM## programs have different expected death and defaulting patterns and should not share a single “one-line” target.

For outpatient therapeutic feeding programs, Sphere standards are commonly used for performance interpretation: recovered >75%, died <10%, and defaulted <15% among discharges. Track additional operational indicators that predict individual failure, including median length of stay, proportion with nutritional oedema at entry, and the fraction referred to inpatient care for complications.

For inpatient services, death thresholds are often expected to be lower than outpatient benchmarks when stabilization is timely and iatrogenic harm is minimized, but local targets must account for referral severity and comorbidity burden. High-yield inpatient quality signals include deaths within the first 48–72 hours (often linked to shock, hypoglycemia, hypothermia, or fluid errors), frequency of inappropriate ##IV## fluid use, and adherence to phased feeding and electrolyte protocols.

Build a shared non-response dashboard across settings. Monitor the proportion failing to regain appetite by day 4 (inpatient), oedema not reducing by day 4 or persisting to day 10, and weight gain <5 g/kg/day after catch-up initiation. When these thresholds rise, treat it as a system problem that triggers protocol audit (therapeutic food preparation, stock-outs, dosing accuracy, referral pathways) rather than blaming individual caregivers. References:

- The Sphere Project, Sphere Handbook (minimum standards in humanitarian response) — https://www.ndsu.edu/sites/default/files/fileadmin/emgt/The_Sphere_Project_Handbook_2011.pdf

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 (NCBI Bookshelf) — https://www.ncbi.nlm.nih.gov/books/NBK601654/
- NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf
- WHO, WFP, UNSCN and UNICEF, Joint statement on community-based management of severe acute malnutrition, 2007 — https://www.unicef.org/media/96981/file/StatementWHOWFP-SCN-andUNICEFonCommunity-Based-Management-of-SAM.pdf

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFANTS UNDER SIX MONTHS AND COMORBIDITY MODIFICATIONS;
SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFANTS UNDER SIX MONTHS AND COMORBIDITY MODIFICATIONS;FEEDING MANAGEMENT IN INFANTS UNDER 6 MONTHS;
Management of acute malnutrition in infants under 6 months is feeding-first and dyad-focused. The therapeutic goal is restoration of effective breastfeeding (or safe alternative feeding) while preventing hypoglycemia, hypothermia, and infection.

Prioritize direct breastfeeding with skilled observation of attachment, positioning, and milk transfer. If the infant is weak, use expressed breast milk by cup/spoon while continuing attempts at direct breastfeeding. When supply is initially insufficient, support relactation and consider the supplementary suckling technique (##SST##), where the infant suckles at the breast while receiving supplemental milk through a thin tube; this increases intake and stimulates milk production.

Use starter therapeutic feeds only as a temporary bridge when breast milk is not yet adequate, and avoid overly concentrated formulas that increase solute load. For non-breastfed infants without prospects of breastfeeding, use program-approved replacement feeding with strict hygiene counseling and close follow-up; treat inability to ensure safe preparation as an admission indication.

Stabilize acute problems using standard ##SAM## safety rules: avoid fasting with frequent small feeds, maintain warmth, treat suspected infection early, and avoid unnecessary ##IV## fluids. Daily reassessment must include feeding effectiveness, urine output proxy, temperature, weight trend, and maternal barriers.

Discharge when feeding is effective on observation, the infant is clinically stable, and there is clear weight gain on the home-feeding plan. Arrange at least weekly follow-up until sustained gain is demonstrated, and provide explicit return-today danger signs (poor feeding, lethargy, fever or hypothermia, fast breathing, vomiting, or new edema).
References:
- World Health Organization, Severe acute malnutrition in infants (ELENA guidance) — https://www.who.int/tools/elena/interventions/sam-infants
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/

- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;INFANTS UNDER SIX MONTHS AND COMORBIDITY MODIFICATIONS;SAM MANAGEMENT MODIFICATIONS IN MAJOR COMORBIDITIES;
Comorbid disease changes risk and modifies the standard ##SAM## pathway. Teach trainees to adjust feeding pace, fluid thresholds, antibiotic choices, monitoring intensity, and discharge planning based on the dominant comorbidity rather than applying a single protocol to all.

##TB## suspected or confirmed: maintain a low threshold for screening in non-responders; do not rely on a negative ##TST##; coordinate diagnostics and follow-up with ##NTEP##. Provide pyridoxine prophylaxis with isoniazid-containing regimens and anticipate slower weight recovery.

##HIV## exposed or infected: continue standard staged feeding and routine potassium/magnesium. Empiric antibiotics remain required even if cotrimoxazole prophylaxis is in place; if severe pneumonia with profound hypoxia suggests ##PCP##, add high-dose cotrimoxazole per protocol. Start ##ART## after initial stabilization of sepsis and metabolic complications, with linkage to the nearest ##ART## center.

Neurodisability or unsafe swallow: treat aspiration risk as a complication. Use slower feeds, head elevation, early ##NG## support with strict placement confirmation, and longer inpatient observation; caregiver skill in safe feeding is a discharge prerequisite.

Congenital heart disease or chronic lung disease: escalate feeds more slowly, restrict additional fluids, and monitor for rising respiratory rate and pulse during feed advancement; seek early cardiology input.

##CKD## or ##CLD##: avoid nephrotoxic antibiotic stacking when alternatives exist, interpret electrolytes cautiously, and reduce or closely monitor phosphate replacement if refeeding syndrome is suspected. Use a lower threshold for referral when close monitoring cannot be ensured.
References:
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- National Tuberculosis Elimination Programme, Paediatric TB management guideline — https://tbcindia-wp.azurewebsites.net/wp-

content/uploads/2023/05/9534339438Paediatric_TB_Mangement_Guideline_22082022_V1.pdf
- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;CASE-BASED MANAGEMENT DECISIONS IN EARLY INPATIENT CARE;
Vignette 1: a 16-month child with visible severe wasting arrives with watery diarrhea, cold hands, capillary refill >3 seconds, and lethargy. A junior resident proposes 20 mL/kg saline boluses for “severe dehydration.” In ##SAM##, this pattern can represent septic shock with unreliable dehydration signs. Correct approach is cautious shock protocol with intensive monitoring, early antibiotics, glucose correction, warming, and slow reassessment-driven rehydration rather than repeated boluses.

Vignette 2: a 2-year child with bilateral pitting edema and dermatosis is prescribed furosemide to “remove fluid.” Nutritional edema reflects potassium depletion and sodium retention, and diuretics are contraindicated. Correct approach is sodium restriction, routine potassium and magnesium, staged feeding, and aggressive search for skin sepsis or pneumonia, adding antistaphylococcal coverage when indicated.

Vignette 3: a non-edematous ##SAM## child is advanced rapidly to high-volume catch-up feeds on day 2 and develops rising respiratory rate and new edema. Interpret this as possible fluid overload or refeeding complication rather than “treatment failure.” Reduce feed advancement toward stabilization volumes, stop extra fluids, correct potassium and magnesium, check phosphate when available, and reassess for concurrent infection. Resume gradual escalation only when vitals stabilize and appetite improves.

Across scenarios, the teaching point is that in ##SAM##, physiologic fragility makes trend monitoring (respiratory rate, pulse, temperature, intake) more reliable than single physical signs, and early “usual pediatrics” interventions can be lethal.
References:
- Ministry of Health and Family Welfare, Operational guidelines on facility based management of children with severe acute malnutrition, 2011 — https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/operational_guidelines_on_fbmc_with_sam.pdf
- World Health Organization, Pocket book of hospital care for children: Severe acute malnutrition chapter — https://www.ncbi.nlm.nih.gov/books/NBK154454/
- National Health Mission, Participant manual for facility based care of severe acute malnutrition — https://nhm.gov.in/images/pdf/programmes/child-health/IECmaterials/PARTICIPANTMANUAL_FBCSA-Malnutrition.pdf
- World Health Organization, Dehydration in severely malnourished children (ELENA guidance) — https://www.who.int/tools/elena/interventions/dehydration-sam

SEVERE ACUTE MALNUTRITION - (SAM) MANAGEMENT;CASE-BASED MANAGEMENT DECISIONS IN DISCHARGE AND OUTPATIENT CARE;
Vignette 1: a child’s ##WHZ## improves to ≥−2, but ##MUAC## remains <125 mm. The team plans discharge because “weight-for-height normalized.” Correct approach is to anchor recovery

to the admission criterion and require repeated edema-free stability; discordance is common and continued therapy with close follow-up is safer than single-metric discharge.

Vignette 2: a 3-month infant with low weight-for-length has weak suck and the mother reports “no milk.” The error is prescribing therapeutic milk without addressing feeding effectiveness. Correct approach is mother–infant assessment, hypoglycemia and hypothermia prevention, urgent lactation support, expressed breast milk plans, and use of ##SST## if needed; disposition depends on observed effective feeding, not anthropometry alone.
Vignette 3: a child enrolled in outpatient ##CMAM## misses two follow-up visits. Recording “defaulter” without action is a safety failure. Correct approach is immediate tracing (home visit or phone contact), assessment for intercurrent illness or social barriers, and urgent facility referral if there is weight loss, edema, or danger signs.

Vignette 4: a child with recurrent infections and wasting presents with severe pneumonia and profound hypoxia; caregiver history suggests possible ##HIV##. Correct approach is to treat as complicated ##SAM## with broad empiric antibiotics, consider ##PCP## and add high-dose cotrimoxazole when indicated, and plan ##HIV## testing and linkage, starting ##ART## after stabilization.

Across scenarios, teach trainees to treat follow-up reliability and caregiver competence as clinical variables that change disposition, dosing decisions, and escalation thresholds.
References:

- World Health Organization, Guideline on the prevention and management of wasting and nutritional oedema, 2023 — https://www.ncbi.nlm.nih.gov/books/NBK601654/ - NITI Aayog, Community-based management of acute malnutrition in India report — https://www.niti.gov.in/sites/default/files/2023-12/CMAM-report-NITI-Aayog.pdf - World Health Organization, Severe acute malnutrition in infants (ELENA guidance) — https://www.who.int/tools/elena/interventions/sam-infants
- Indian Academy of Pediatrics, Standard Treatment Guidelines: Severe acute malnutrition, 2022 — https://iapindia.org/pdf/Ch-136-Severe-Acute-Malnutrition.pdf